

# BAAZ & UDAY JKBOSE

2026 CLASS 11<sup>TH</sup>

## OSCILLATIONS

LECTURE-02 | PHYSICS

WAJAHID SIR (Gabbar Sir)



# Topics To Be Covered

1 Revision dec-①

2 Graphs  $x_t$  |  $v_t$  |  $a_t$

3 Spring

4 Simple Pendulum

5 Energy in SHM

# Recap - 1

⇒ Motion.

⇒ Translation

⇒ Periodic

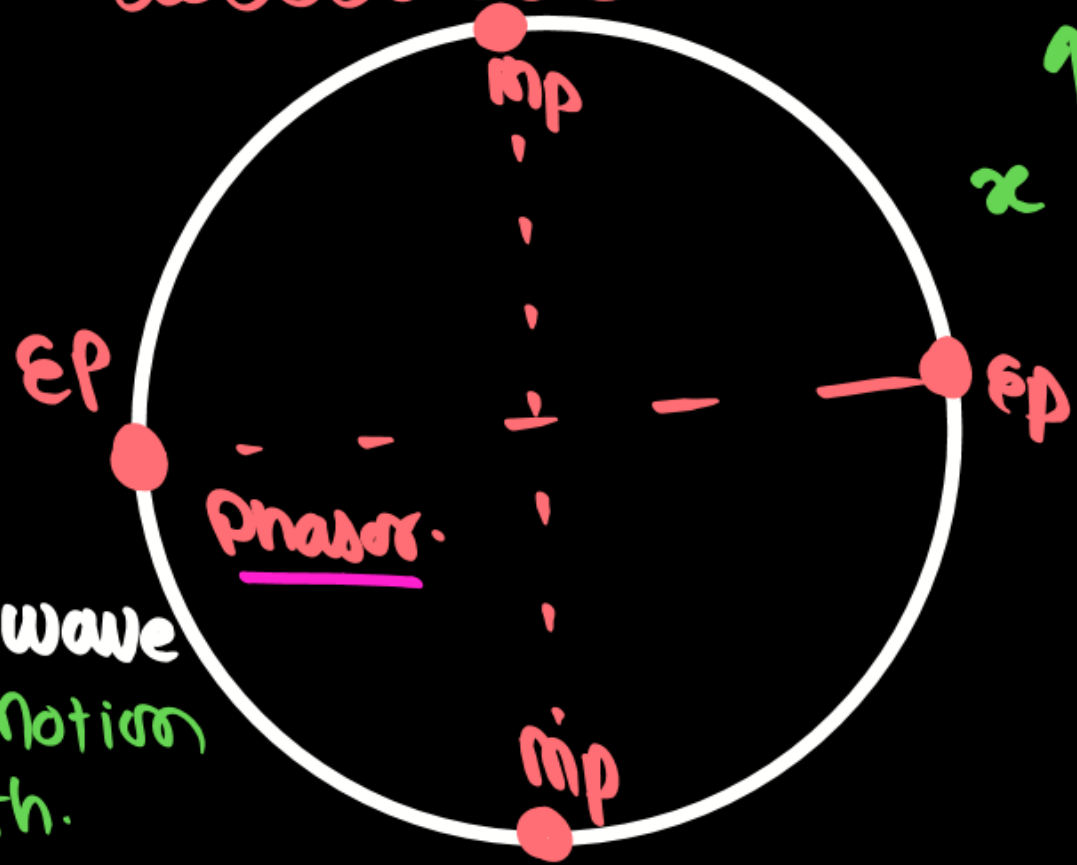
✓ Revolution

✓ Rotation.

\* ✓ Oscillation / wave  
Vibrational motion  
To and forth.



Simple harmonic motion



$F \propto -x$

Imp

Derive

①  $x = A \sin(\omega t + \phi)$

②  $v = A\omega \cos(\omega t + \phi)$

③  $a = -A\omega^2 \sin(\omega t + \phi)$

↓

$a = -\omega^2 x$

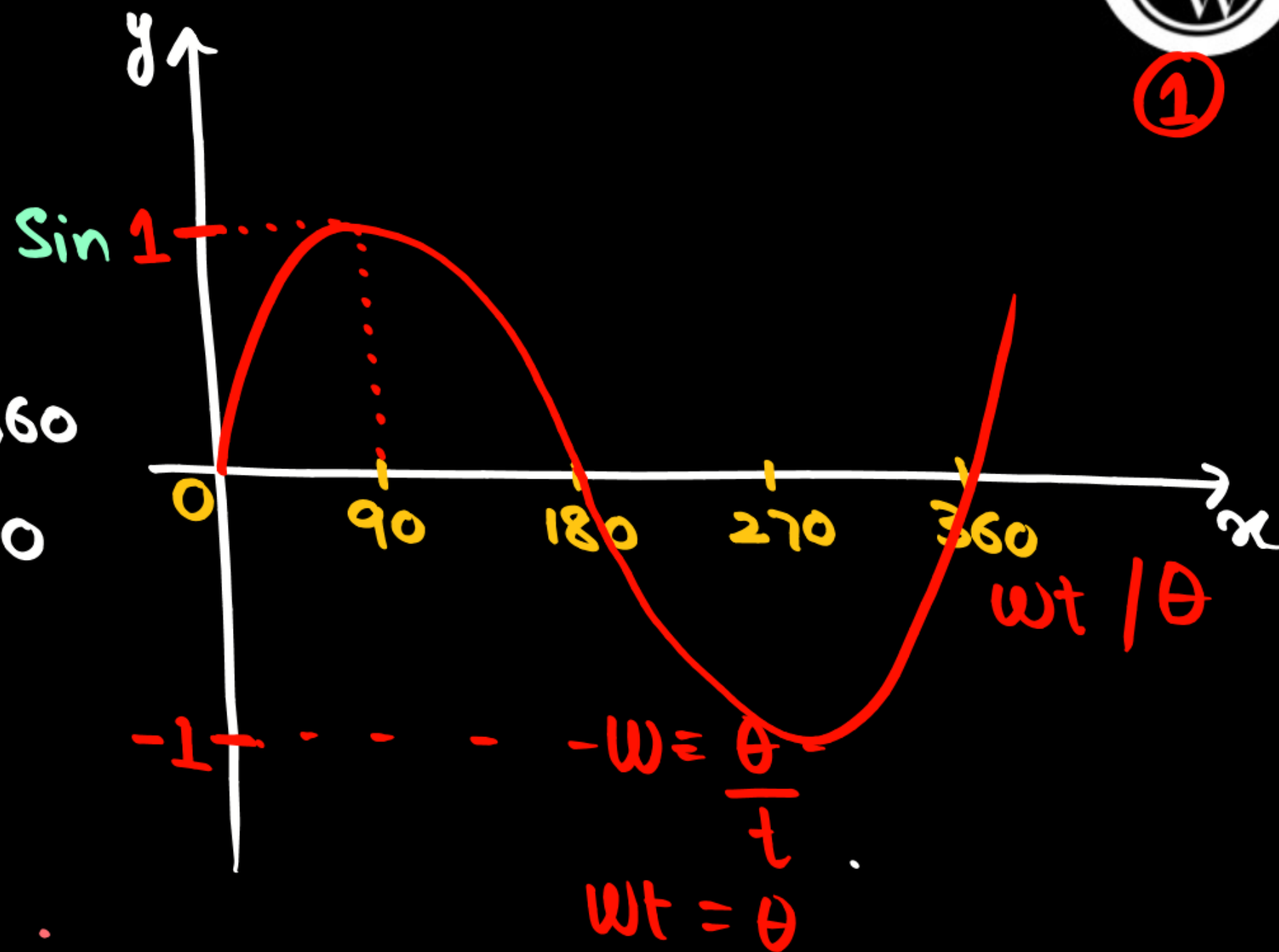
④  $F = -m\omega^2 x$

## lecture no. 2

Basics

$\Rightarrow$  Sin graph

|     |           |     |      |      |      |
|-----|-----------|-----|------|------|------|
|     | <u>0°</u> | 90° | 180° | 270° | 360° |
| Sin | <u>0</u>  | 1   | 0    | -1   | 0    |

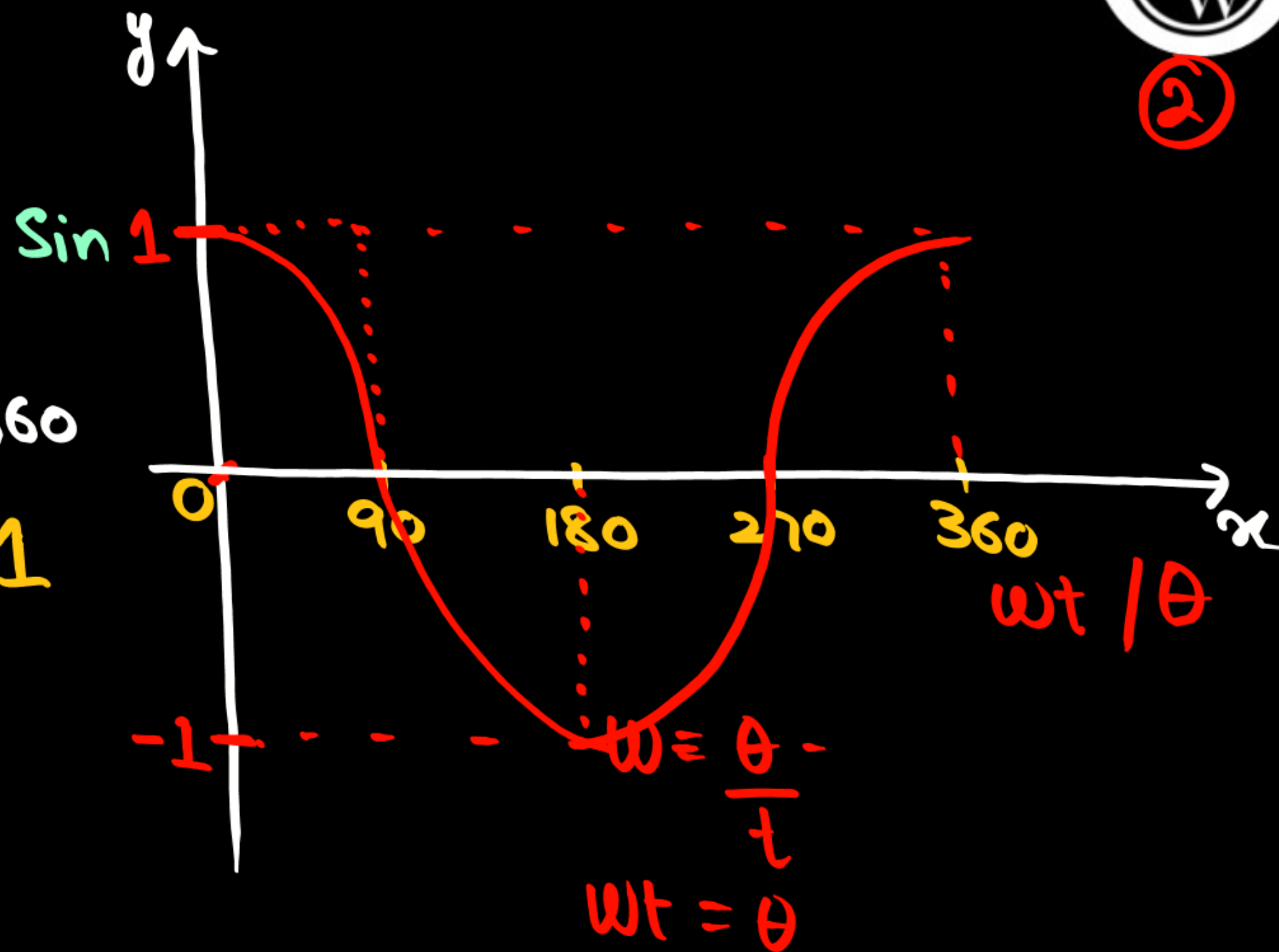


①

Basics

$\Rightarrow$  cos graph

|            |          |     |             |            |          |
|------------|----------|-----|-------------|------------|----------|
|            | <u>0</u> | 90° | <u>180°</u> | <u>270</u> | 360      |
| <u>cos</u> | <u>1</u> | 0   | <u>-1</u>   | <u>0</u>   | <u>1</u> |



# Displacement time graph in SHM wave

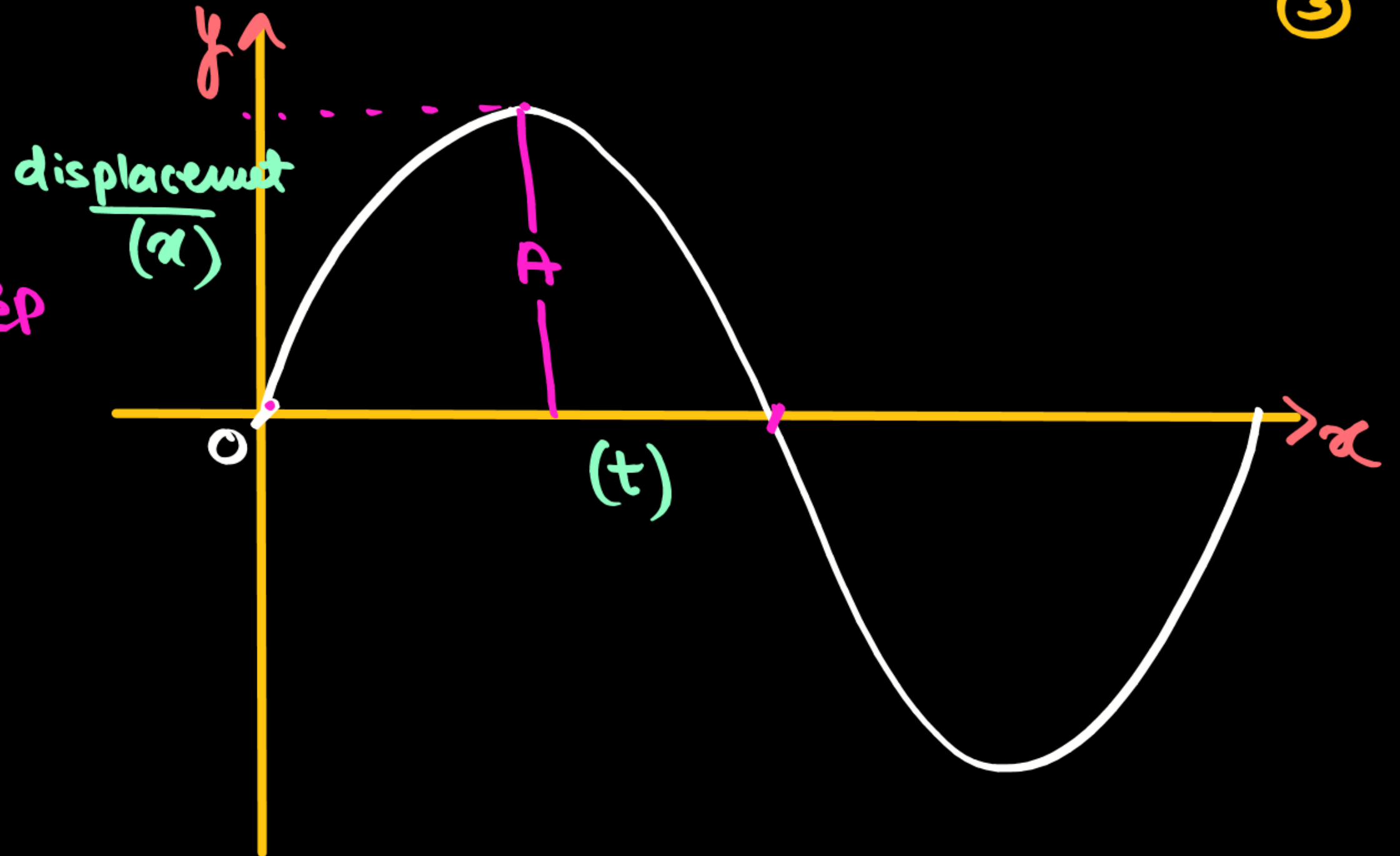
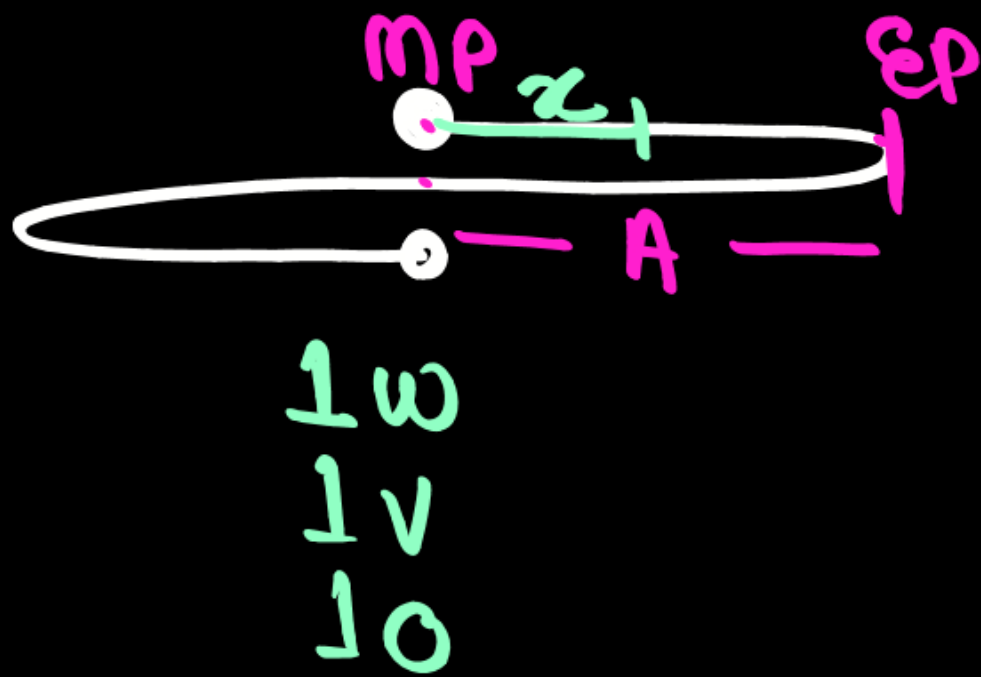
2021



3

$$x = A \sin(\omega t)$$

$$x = A \sin \theta$$



2024

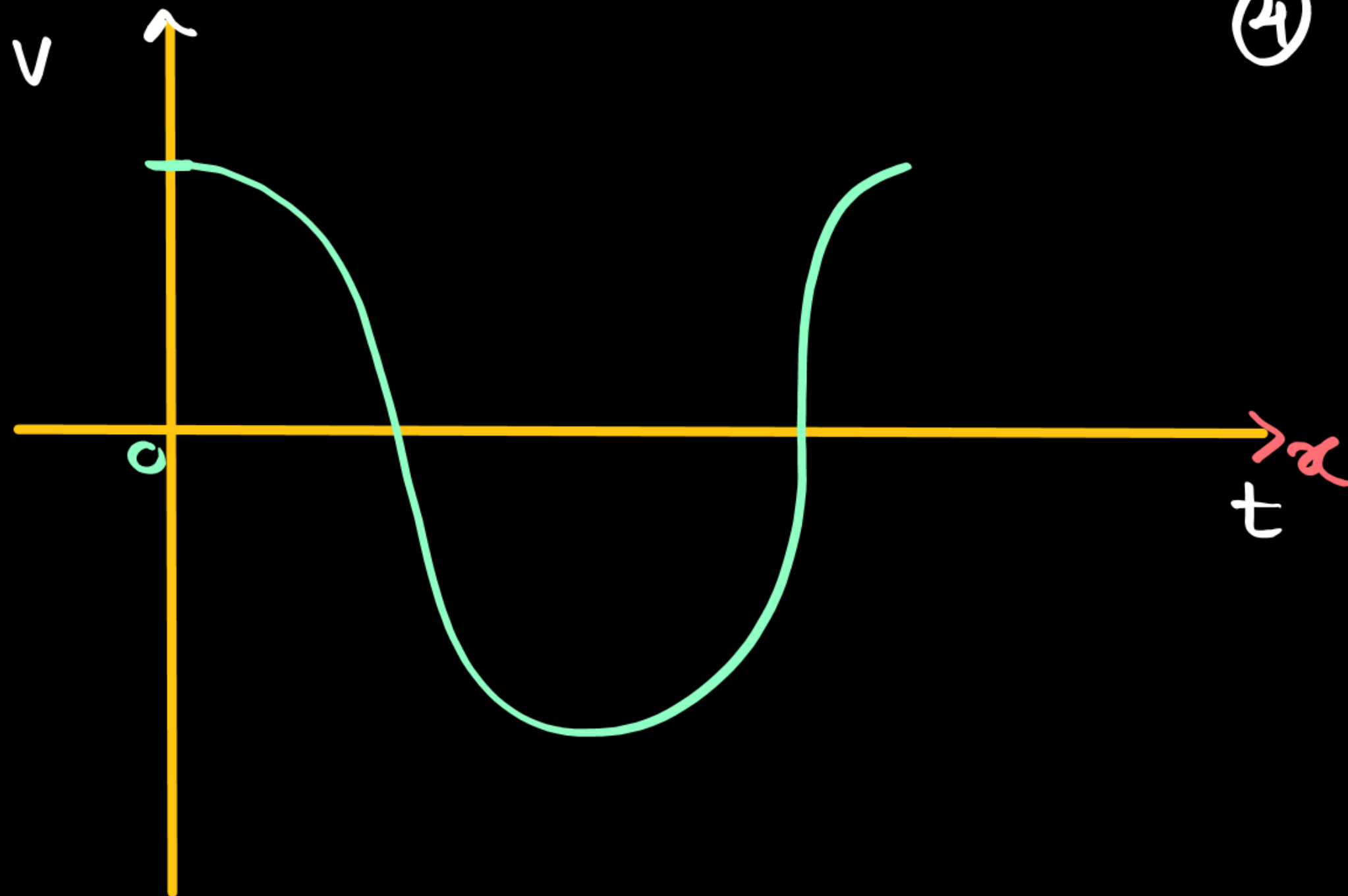
# Velocity — time graph in SHM <sup>wave</sup>

$$v = A\omega \cos \omega t$$

$$v = A\omega \cos \theta$$

$$\omega = \frac{\theta}{t}$$

$$\omega t = \theta$$



at

Acceleration — time graph in SHM wave

2025✓



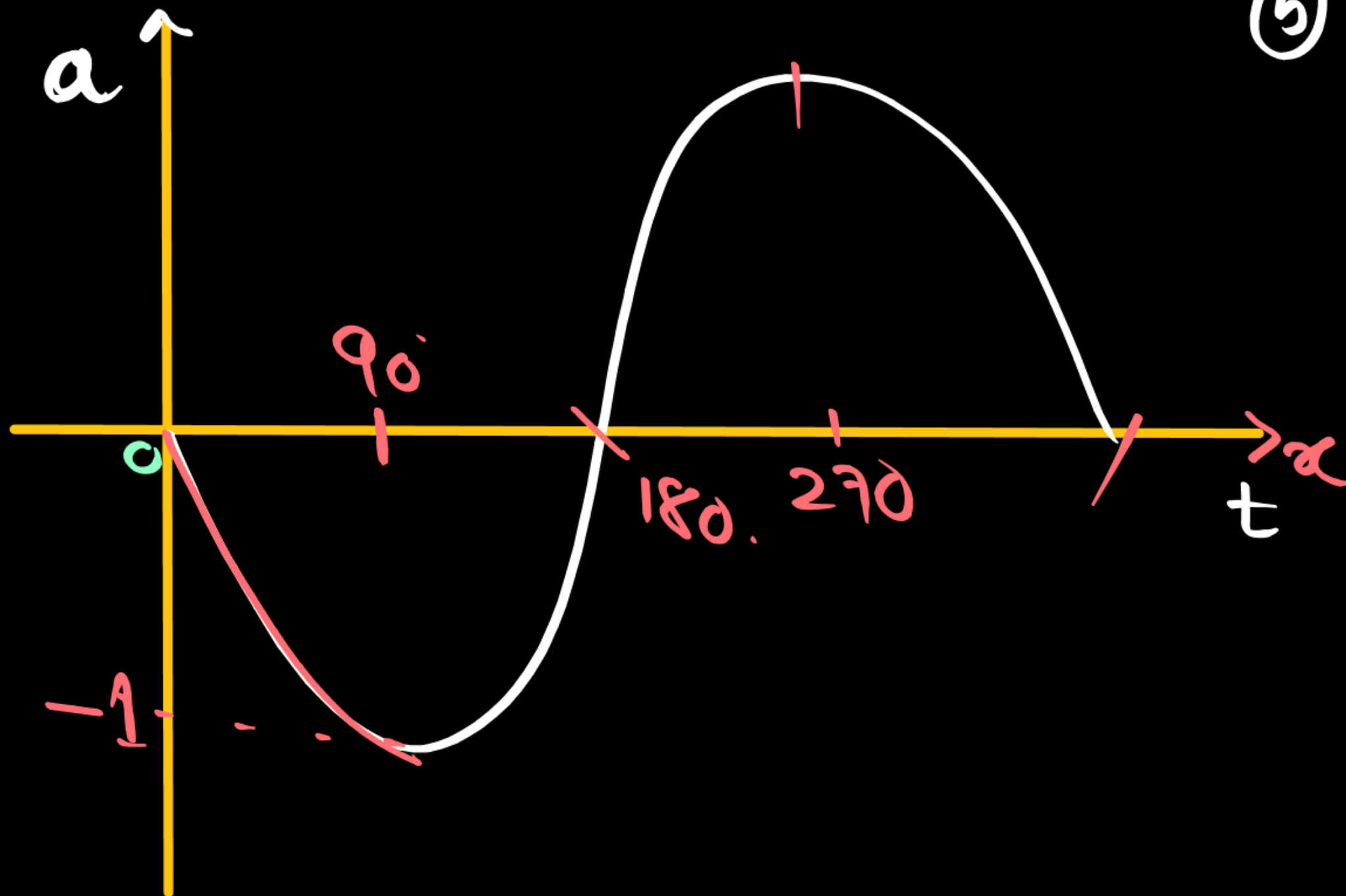
5

$$a = -A\omega^2 \sin(\omega t)$$

$$a = A\omega^2 \underline{-\sin(\omega t)}$$

$$\sin(90) = 1$$

$$-\sin(90) = -1$$



Example of SHM

$$F_{SHM} = -m\omega^2 x$$

Spring

Simple pendulum

✓ Derivation

$\omega$

T ✓ (2021)

✓ Derivation

$\omega$

T (5 mark) 2023)

## Spring

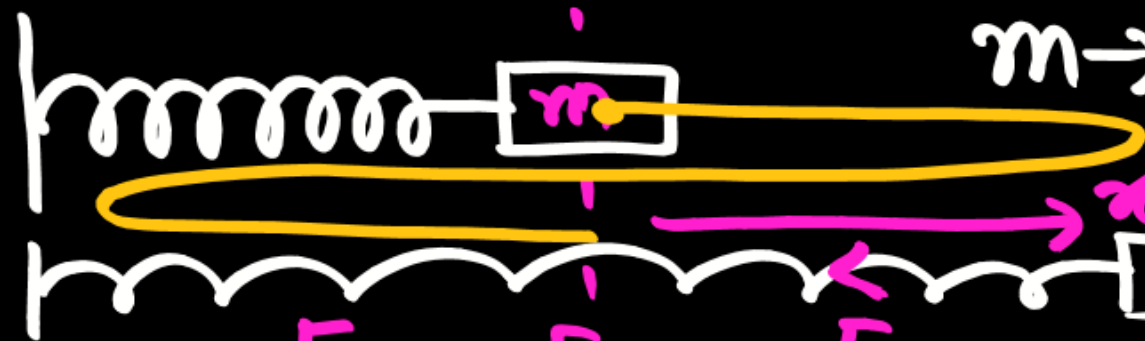
$\omega \rightarrow$  Angular frequency

$T \rightarrow$  Time period of Spring

The Spring motion is  
an Example of  
SHM

$k \rightarrow$  Spring  
constant

$m \rightarrow$  mass  
of object



$$F_s = F_{SHM}$$

$$+kx = +m\omega^2 x$$

$$k = m\omega^2$$

$$k = m\omega^2$$

$$\frac{k}{m} = \omega^2$$

Angular

$$\sqrt{\frac{k}{m}} = \omega$$

$$\omega = \frac{\theta}{t} = \frac{2\pi}{T}$$

$$\omega = \frac{2\pi}{T}$$

$$\sqrt{\frac{k}{m}} = \frac{2\pi}{T}$$

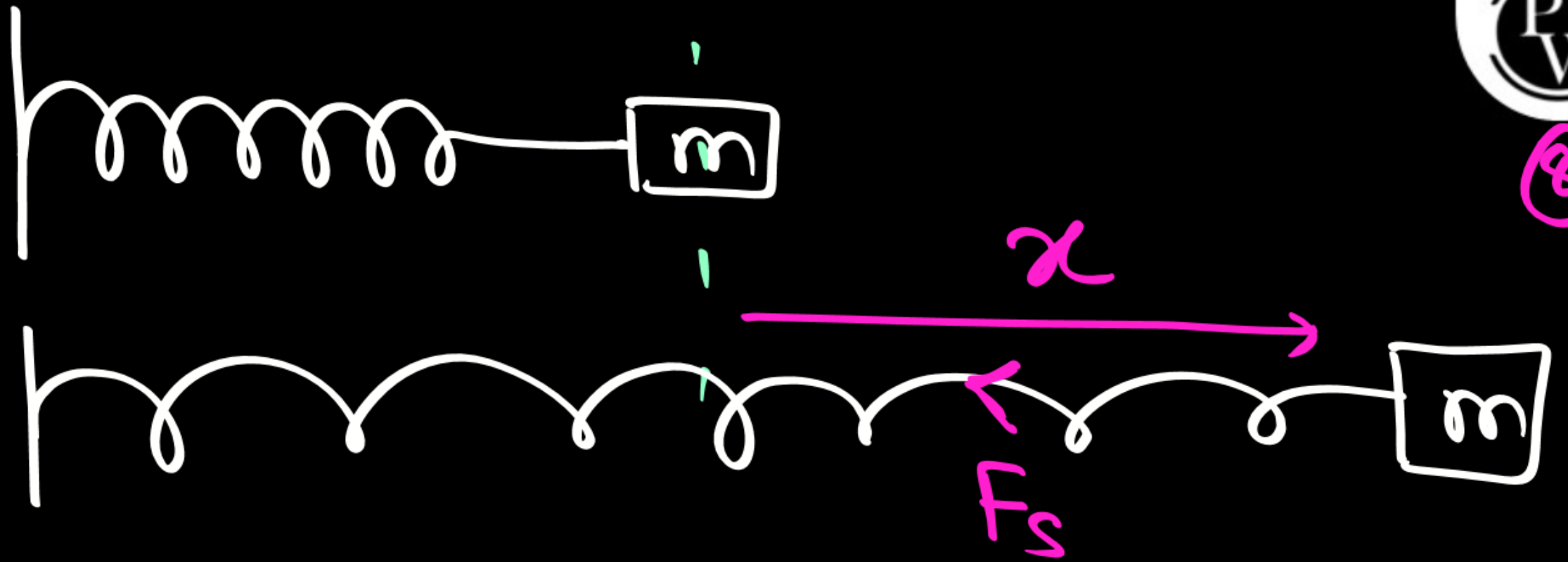
$$\sqrt{\frac{m}{k}} = \frac{T}{2\pi}$$

$$\frac{2\pi}{\sqrt{\frac{k}{m}}} = T$$



7

Note

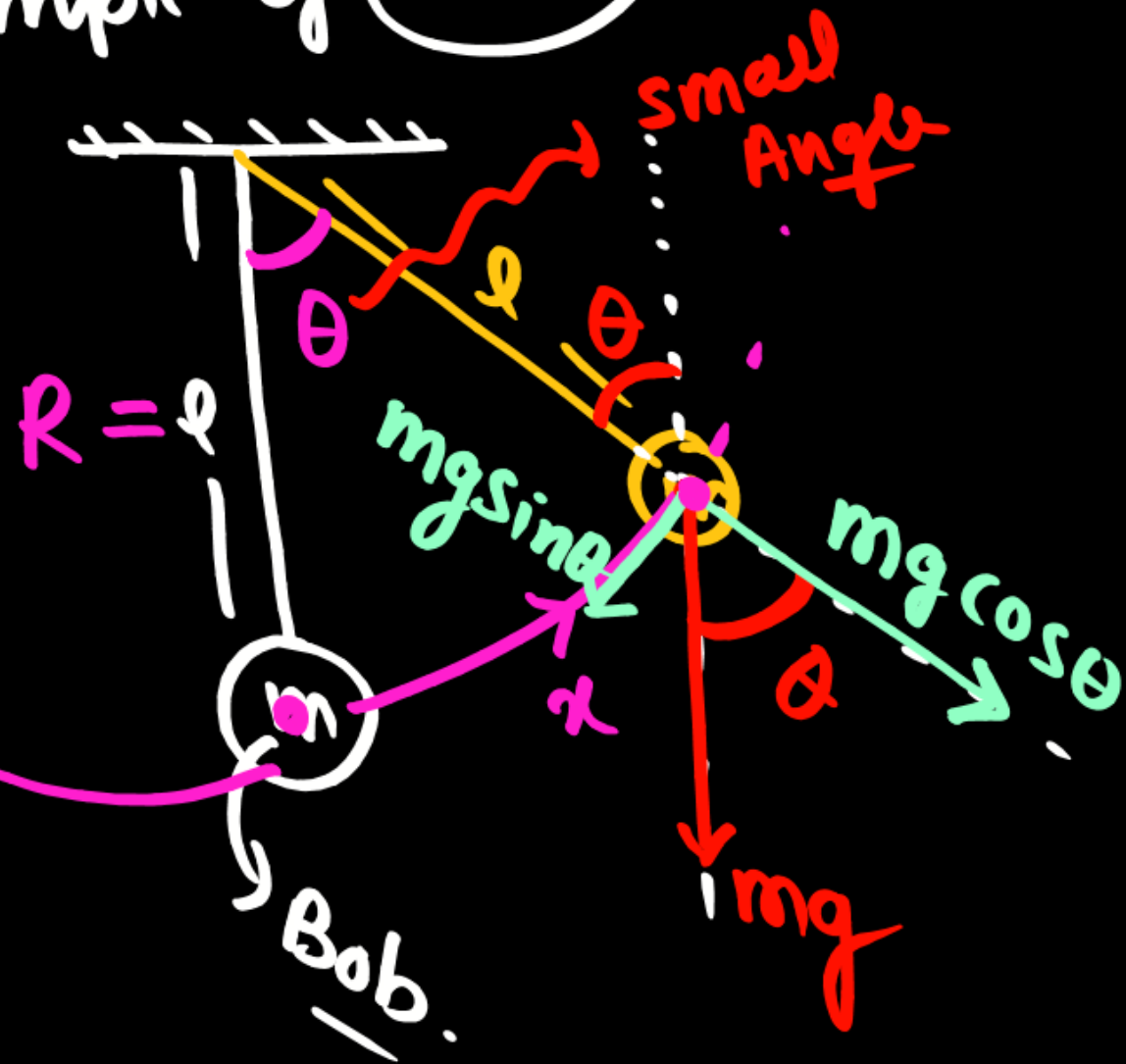


$$F \propto -x$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

# Simple pendulum

The motion in Simple Pendulum is also the Example of SHM



$$F_{SHM} = -mg \sin \theta \checkmark$$

For Small Angle Approx.

$$\sin \theta = \theta \quad (10)$$

$$F_{SHM} = -mg \theta$$

$$F_{SHM} = -mg \frac{x}{l}$$

$$+m\omega^2 x = +mg \frac{x}{l}$$

$$\omega^2 = g/l$$

Angular frequency

$$\omega = \sqrt{g/l}$$

$$\theta = \frac{\text{Arc}}{R}$$

$$\theta = \frac{x}{l}$$

$$\omega = \sqrt{\frac{g}{l}}$$

$$\omega = \frac{2\pi}{T}$$

$$\uparrow \uparrow T = 2\pi \sqrt{\frac{l}{g}} \uparrow$$



$$\left(\frac{2\pi}{T}\right) = \left(\sqrt{\frac{g}{l}}\right)$$

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

using concept of SHM

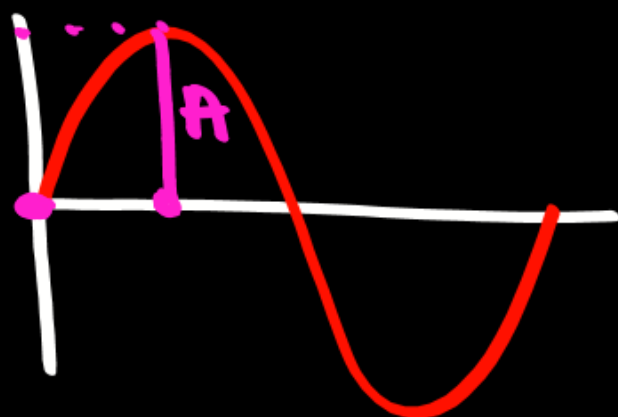
Note  
2023  
MCQ

No the time period of Simple pendulum donot depend on mass ✓

(Not e.)

## Wave length $\lambda$

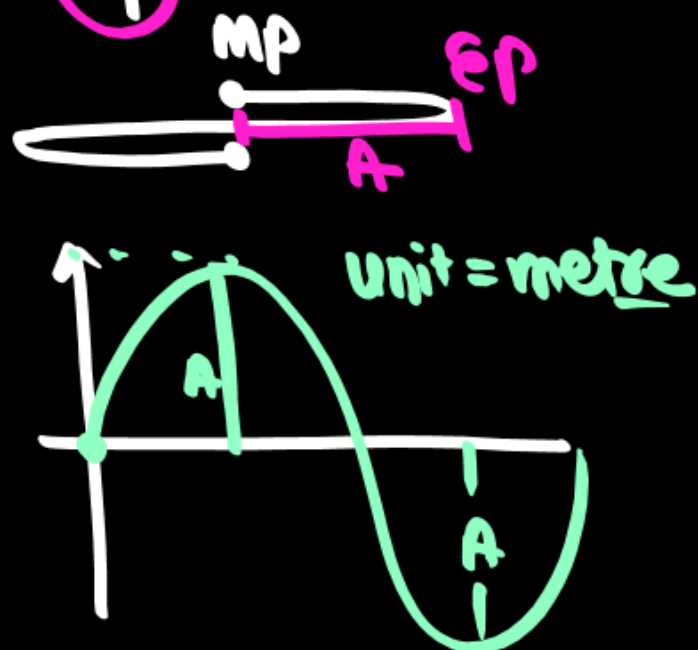
The length of full wave



unit = metre  
↳ 2019 ✓

## Amplitude

The distance b/w  $mp$  and  $ep$



unit = metre

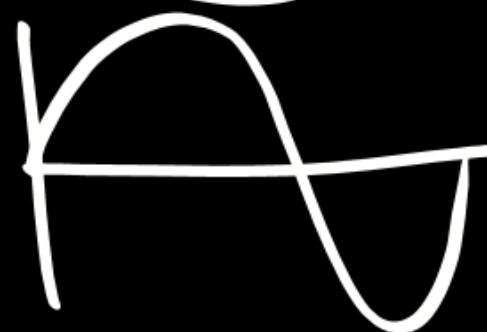
$\omega \rightarrow$  Angular velocity  
Angular frequency

## Time period

Time to complete 1 oscillation



unit = sec

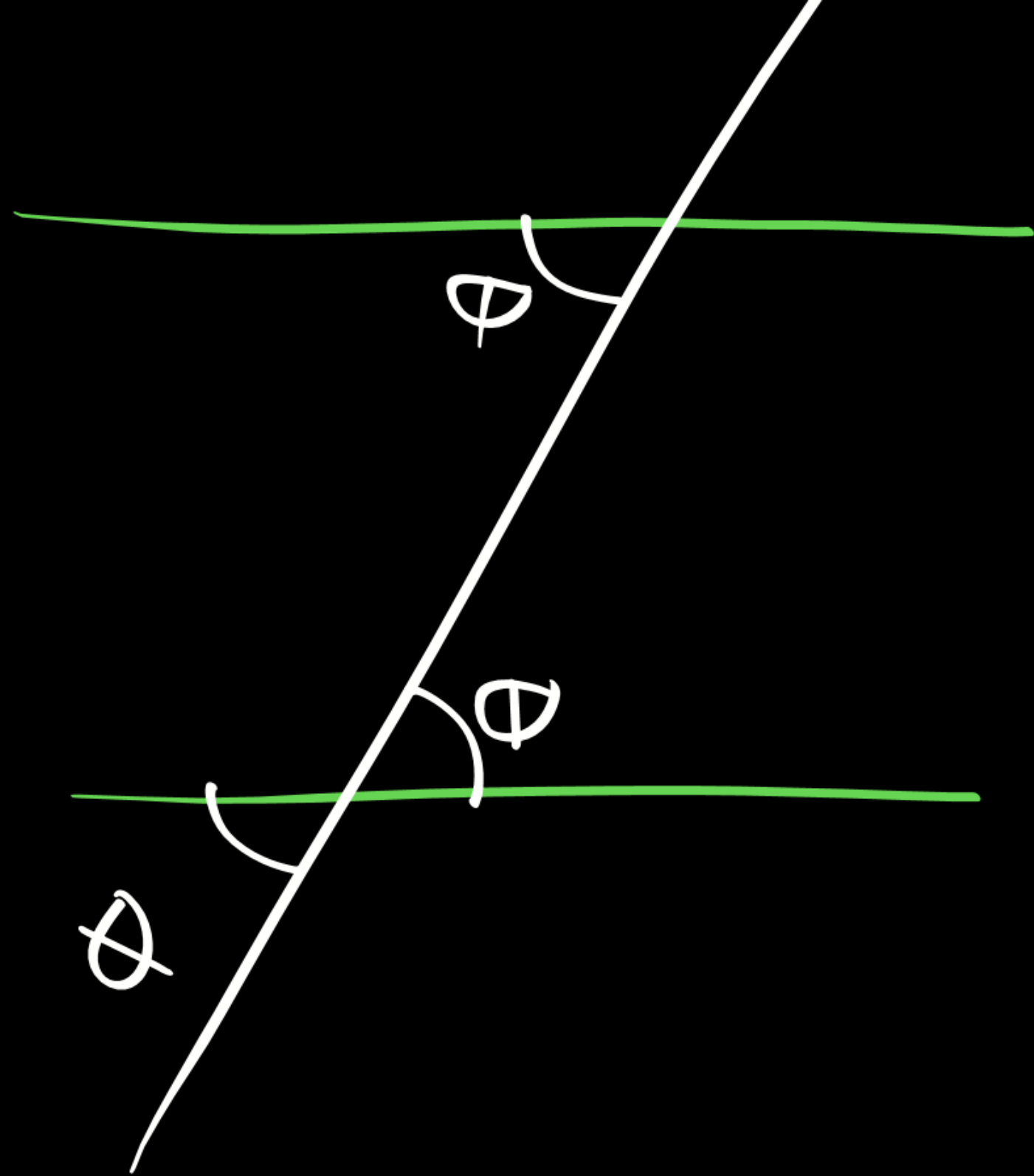


## Frequency $\gamma$

Reciprocal of time period

$$\gamma = \frac{1}{T}$$

$$\gamma = \frac{1}{T} = \frac{1}{\text{Sec}} = \text{Hertz}$$



# SUMMARY



Sim ✓

$\lambda$

cos ✓

A

$x_t$

T

$v_t$

$\vec{v}$

$a_t$

$\omega$

SP

Spring

# Second Pendulum

H/W



Mehnat Karo Mehnat karo  
Mehnat karne se ho jata hai  
Insallah apke <sup>so</sup> ~~498~~ No. Aayenge  
Bas Yhi motivation apne dil me  
rkho bs



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60 day ✓

1 chapter X

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LW+BC X

**Thank**  
*You*



Pala Wallah